Loacker

Digitizing the hazelnut value chain

A deep-dive into how Loacker drives quality hazelnut farming and builds a transparent farm monitoring process

Cropin

Contents

Loacker: A history as rich as its products

Loacker, an Italian company based in South Tyrol (Italy), produces wafers, chocolates, and derivative products with the promise of wholesome natural goodness, highest quality standards and absolute deliciousness. It was originally founded by Alfons Loacker in 1925 as a pastry shop in Bolzano, Italy. The brand is well known for its wafer and chocolate specialties, particularly the 'Classic Napolitaner' product line, consisting of three fragrant wafers and two rich layers of exquisite hazelnut cream filling (which accounts for 75% of the product). All Loacker products are manufactured in the heart of the Alps, in Auna di Sotto (South Tyrol) and Heinfels (East Tyrol), using the most modern processes. With production facilities in Bolzano, Ritten, and Heinfels (Austria) and employing more than 4,000 people, the company gives supreme importance to selected, natural ingredients. And its products are completely manufactured with no added flavorings, colors, or preservatives. A global player with a presence in over 100 countries, Loacker's primary market is Italy, while other big markets include Saudi Arabia, Israel, USA and China.

"Loacker Progetto Noccioleti Italiani": A product of sustainability and a love for nature

Loacker initiated a long-term new project, "Loacker Progetto Noccioleti Italiani", to source 100% sustainably produced Italian hazelnuts. Under this program, the company aims to own the production of hazelnuts in Tuscany and source through long-term contract farming partnerships in four Italian regions. This project assures end-to-end traceability of the Italian hazelnuts and the complete control over quality and agronomic techniques in hazelnut orchards. Under the initiative, Loacker covered 1,30,000 hazelnut plants across 360 hectares (nearly 890 acres) in Italy. The project culminated in 80 regional grower partnerships in six regions. Every contract signed with the farmers also included an agronomic protocol covering sustainable practices that go over national legal standards in hazelnut cultivation.

The challenge - perfectly balancing our contract and farmer collaboration

Loacker kick-started the "Italian hazelnut groves" project focusing on three elements: **Quality, Italian Origin,** and **Sustainability**. The company invested in 240 hectares having two self-owned estates, namely Corte Migliorina and Collelungo, located in the province of Grosseto. In addition, Loacker started a contract farming program that also hoped to provide a concrete income alternative and diversification possibility to farmers. The hazelnut project targeted complete control over the hazelnut chain in terms of traceability, processing and quality. Soon enough, the brand came face-to-face with numerous tangible and intangible challenges, such as:

- The inability to provide technical advice to farmers and monitor farms.
- Poor visibility into farmers' cultivation processes and demarcation of their plots.
- Inherent quality issues of the hazelnut crop. While hazelnut is among the most sustainable crops requiring few interventions during cultivation, infestations of the stink bug from the North to the South can destroy the fruit and, therefore, must be tackled.

The goals in the journey to ensure goodness and pure joy of ultimate taste

Loacker was looking to invest in a simple, tech-enabled solution that would help achieve the following goals:

Goal 1:

Farmer digitization to assure transparency and sustainability in hazelnut farming

To ensure the adoption of environmentally and socially sustainable farming practices, which are central to the brand image of Loacker, the digitization of hazelnut farming became crucial. For this, Loacker had to:

- Nurture and grow the agronomic skills of farmers necessary for sustainable management of the hazelnut crop via a centralized platform that ensures a good relationship
- Closely monitor farm activities and treatments to ensure they're done at the right time, using intuitive digital tools, thereby walking every step with the farmer

Goal 2:

Crop and warehouse management to ensure traceability in hazelnut production

Crop and warehouse management is essential in achieving farm-to-fork traceability. This is especially crucial for Loacker, which has always maintained the use of feedstock of national origin and highlighted the same in its labeling. Thus, Loacker required a digitization solution for:

- Complete visibility into the activities at its chocolate processing center and warehouses – right from cleaning, drying, quality control, and packaging to the creation of the final end-product, the chocolate
- 360-degree traceability of hazelnut and management of goods receipt from the warehouse to the final destination

Goal 3:

Close monitoring of activities and treatments

The final goal entailed increased transparency and ensuring sustainability in activities. This necessitated an understanding of key concerns such as:

- How much money is invested in ensuring quality harvest produce and reviewing who did the transaction and for which farmer
- Keeping track of receipts generated in each phase of the transaction.

The Results: Cropin enables Loacker to keep its promise of quality in every bite

It was clear that digitalization of the whole supply chain would be essential for Loacker. The company selected Cropin as the best suitable partner to take the process forward. Cropin's farm digitization and traceability apps, **Cropin Grow (SmartFarm)** and **Cropin Trace (RootTrace)**, were offered both in English and Italian to support Loacker to easily achieve its stated objectives of ensuring quality, Italian origin, and sustainability in hazelnut production. The flexibility of Cropin's solution goes a long way in the hazelnut project's success.

Ag-cloud technologies to the rescue:

A. Cropin Grow (SmartFarm) - Farm management solution

Cropin's end-to-end farm management solution utilizes ground data from farms, and weather warnings, along with precise and deep understanding from satellite monitoring to help data-driven farm and business management.

- Cropin's solutions are user-friendly and easy to handle after some training sessions
- The reports provided by Cropin, are crucial to conducting the entire quality check and assurance processes along the supply chain.

Key features:

Geotag farm plots and detect crops

Monitor input usage

Share weather-related advisory

Monitor crop health remotely

Set pest and disease alerts

Manage alerts and activity log

Set up workflows and tasks for the field team

Customizable business intelligence dashboard and reporting

The scheduled activities planned with the platform for Loacker are:

Spring fertilization localized on the row

Foliar fertilization of macro-and micronutrients

Seasonal activities like spring-summer tillage and winter pruning for sapling breeding and bush breeding

Individual foliar fertilization of micronutrients and that of macronutrients

Phytosanitary treatment for leaf fall and breaking buds

Soil preparation for harvesting

Soil enrichment tasks such as green manure sowing and organic fertilization

Shredding - a waste management process to control environmental consequences

Benefits:

End-to-end farm digitization

Improved data capture accuracy

Traceability and output predictability

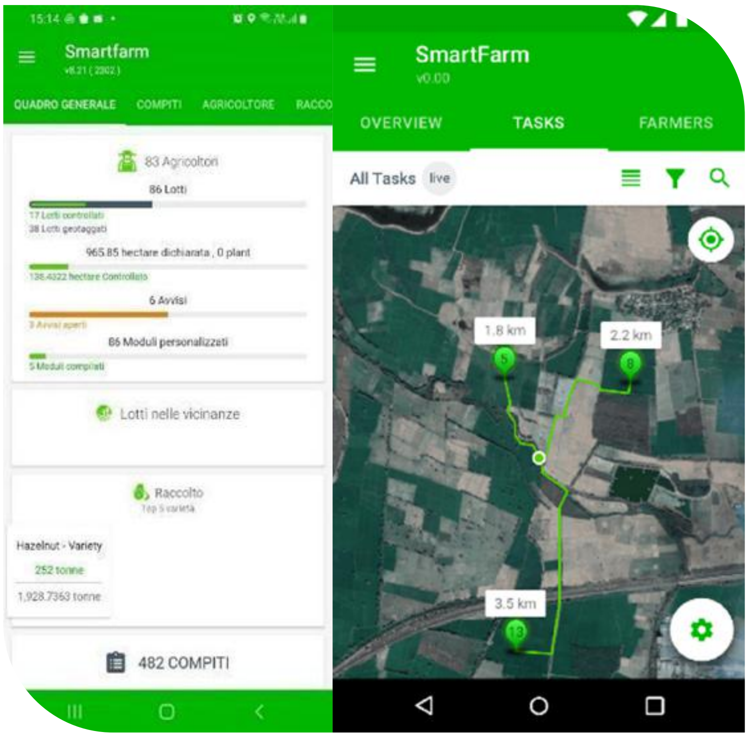
Seamless monitoring and management of farms

Data-driven and agile decision-making that leads to improved farm productivity and profitability

Minimized errors and increased time efficiency

B. Cropin Trace (RootTrace) - Farm-to-fork traceability solution

A comprehensive solution for packhouse, processing, and export companies offering farm-to-fork traceability, compliance, quality control, and flexible inventory management.



The supply chain traceability process starts with quality control and shelling of hazelnuts to the packaging of the shelled nuts. Farms and Collection Centers are located in four different Italian regions of **Veneto, Tuscany, Umbria,** and **Marche**, and there is one processing center in South Tyrol.

Key features:

Complete inventory control with monitoring of warehouse operations

Batch number based unique identification of items

Audit forms to capture quality parameters while in storage

Streamlined order management and fulfilment

Tamper-proof, anti-counterfeit labels

Map to different markets and compliances

Benefits:

Better visibility of inventory and processes

Maximized productivity and profitability

Elimination of stocking and wastage challenges

Zero errors while fulfilling orders

Enhanced product quality and food safety as well as higher transparency leading to increased credibility

Cost and time savings with prompt recalls

By using the two solutions, Loacker ensured transparency in crucial post-harvest and processing activities listed below.

1. Harvest delivery

The farmers bring the harvest, after five years of planting, during August-September. The harvest is not done just once but 2 to 3 times to collect all the hazelnuts. Therefore, in different collection centers, each farmer delivers at least 2-3 lots per year.

2: Cropin Grow (SmartFarm)

A Loacker technician conducts all the weighing, sampling, and quality checks in the collection centers, after which nuts are segregated into Quality A and Quality B using SmartFarm.

3: Price determination

The farmer's price is defined as per the sample submitted in each lot, which they deliver to the collection centers.

4: Transportation

Post collection of hazelnuts at the collection centers, different lots of hazelnuts belonging to multiple farmers are mixed and transported in a truck to the processing center, where nuts are dried and cleaned.

5: Processing

The hazelnuts need to be brought to 6% humidity to stabilize them for storage in the shell. In the months post-harvest, all the hazelnuts are de-shelled, filled into big bags and transferred to the Loacker plants. Final roasting for Loacker's semi-finished products is then carried out in the processing plant in South Tyrol. The products of various hubs are concentrated to create a completely circular economy by enhancing shells to produce thermal energy and the seed cuticle to reuse fibres and proteins by the nutraceutical segment.

Leveraging technology to improve smallholder farmer livelihoods

By deploying farm digitization and supply chain management solutions, Loacker reaffirmed its commitment to traceability and sustainability, and helped improve the livelihood of smallholder farmers.

The impact metrics of the implementation are:

85+ Total Number of Farmers	**95+** Total Number of Plots
225+ Acres of audited Area	**2000+** Tonnes of harvest

Lessons to take forward

The use of digital tools for the "Italian hazelnuts" project is strategic for Loacker - it guarantees the total traceability of the product, from the field to the final consumer. Above all, it empowers the brand to be consistent in its commitment to promoting an inclusive and sustainable economy and driving responsible business management.

This study has demonstrated that a controlled supply chain and a defined agronomic protocol fueled by digital technology can lead to streamlined hazelnut production. To establish–and follow–agronomic technical protocol, hazelnut producers can count on constant comparison with Loacker's agronomists and on the Cropin application, which can be easily consulted on smartphones. In a georeferenced manner, the app can forecast the age of hazelnut collected for each particle and each variety to achieve the goal of complete tracing from the field to the final product (the cream of the wafers).

With assistance from Cropin's project team, Loacker plans to extend the solution to other crops, wherein it will initiate new projects in the future.

About Cropin - A full-stack agri-tech platform

Founded in 2010, Cropin is a pioneer in the Agtech space, having built the first global Intelligent Agriculture Cloud. Cropin's platform enables various stakeholders in the agri-ecosystem to leverage digitization and AI at scale to make decisions that increase efficiency, scale productivity, and strengthen sustainability. Cropin has worked with 250+ customers and has digitized 16 million acres of farmland, improving the livelihoods of more than 7 million farmers. It has built the world's largest and most diverse farming data insights over a decade, spearheading a global 'Ag-intelligence' movement with a knowledge of 488 crops, and 10000 crop varieties in 56 countries. With its AI/ML platform tailor-made for the agriculture ecosystem, Cropin has computed 0.2 billion acres of farmlands across the globe.